

AI Assisted Coding

Assignment-8.1

HTNO: 2303A51236

Batch: 18

Task Description #1

(Password Strength Validator – Apply AI in Security Context)

- Task: Apply AI to generate at least 3 assert test cases for `is_strong_password(password)` and implement the validator function.

- Requirements:

- o Password must have at least 8 characters.

- o Must include uppercase, lowercase, digit, and special character.

- o Must not contain spaces.

Example Assert Test Cases:

```
assert is_strong_password("Abcd@123") == True
```

```
assert is_strong_password("abcd123") == False
```

```
assert is_strong_password("ABCD@1234") == True
```

Code:

```
Lab_assign-8.1.py > ...
1  """Lab 8: Test-Driven Development with AI"""
2  from typing import Dict
3
4  # TASK 1: Password Strength Validator | Req: 8+ chars, uppercase, lowercase, digit, special, no spaces
5  # assert is_strong_password("Abcd@123")==True; assert is_strong_password("abcd123")==False
6  def is_strong_password(p:str)->bool:
7      return isinstance(p,str) and len(p)>=8 and ' ' not in p and any(c.isupper() for c in p) and any(c.islower() for c in p) and any(c.
8          isdigit() for c in p) and any(c in '!@#%^&*()_+=[]{};'\",./<>?\\|`~' for c in p)
9
10 # TASK 2: Number Classification | Req: Classify as Positive/Negative/Zero, handle invalid inputs
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```

```
assignment_8.py > classify_number
7     return isinstance(p,str) and len(p)>=8 and ' ' not in p and any(c.isupper() for c in p)
8
9     # TASK 2: Number Classification | Req: Classify as Positive/Negative/Zero, handle invalid input
10    # assert classify_number(10)=="Positive"; assert classify_number(-5)=="Negative"; assert classify_number(0)=="Zero"
11    def classify_number(n)->str:
12        if n is None or isinstance(n,(str,bool)):return "Invalid"
13        try:num=float(n) if not isinstance(n,(int,float)) else n
14        except:return "Invalid"
15        return "Positive" if num>0 else ("Negative" if num<0 else "Zero")
16
17    # TASK 3: Anagram Checker | Req: Ignore case/spaces/punctuation, handle edge cases
18    # assert is_anagram("listen","silent")==True; assert is_anagram("hello","world")==False
19    def is_anagram(s1:str,s2:str)->bool:
20        if not isinstance(s1,str) or not isinstance(s2,str):return False
21        c1=''.join(x.lower() for x in s1 if x.isalnum())
22        c2=''.join(x.lower() for x in s2 if x.isalnum())
23        return (len(c1)==0 and len(c2)==0) or (len(c1)>0 and len(c2)>0 and sorted(c1)==sorted(c2))

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Enter your choice (1-7): 2
Enter a number: 3456789
Number classification: Positive

Enter your choice (1-7): 2
Enter a number: -23456
Number classification: Negative

Enter your choice (1-7): 2
Enter a number: 000000000000
Number classification: Zero
```

Task Description #3 (Anagram Checker – Apply AI for String Analysis)

- Task: Use AI to generate at least 3 assert test cases for `is_anagram(str1, str2)` and implement the function.
- Requirements:
 - o Ignore case, spaces, and punctuation.
 - o Handle edge cases (empty strings, identical words).

Example Assert Test Cases:

`assert is_anagram("listen", "silent") == True`

```
assert is_anagram("Dormitory", "Dirty Room") == True
```

```

assignment_8.py > classmy_number
17 # TASK 3: Anagram Checker | Req: Ignore case/spaces/punctuation, handle edge cases
18 # assert is_anagram("listen","silent")==True; assert is_anagram("hello","world")==False
19 def is_anagram(s1:str,s2:str)->bool:
20     if not isinstance(s1,str) or not isinstance(s2,str):return False
21     c1=''.join(x.lower() for x in s1 if x.isalnum())
22     c2=''.join(x.lower() for x in s2 if x.isalnum())
23     return (len(c1)==0 and len(c2)==0) or (len(c1)>0 and len(c2)>0 and sorted(c1)==sorted(c2))
24
25 # TASK 4: Inventory Class | Methods: add_item(name, qty), remove_item(name, qty), get_stock()
26 # inv.add_item("Pen",10); assert inv.get_stock("Pen")==10; inv.remove_item("Pen",5); assert inv.get_stock("Pen")==5
27 class Inventory:
28     def __init__(self):self.stock:Dict[str,int]={}
29     def add_item(self,n:str,q:int)->None:
30         if not isinstance(n,str) or not isinstance(q,int) or isinstance(q,bool):raise TypeError
31         if q<0:raise ValueError("Quantity cannot be negative")
32         self.stock[n]=self.stock.get(n,0)+q
33     def remove_item(self,n:str,q:int)->None:
34         if n not in self.stock:raise ValueError("Item not in inventory")
35         if q>self.stock[n]:raise ValueError("Quantity cannot be more than stock")
36         self.stock[n]-=q
37         if self.stock[n]==0:del self.stock[n]
38
39 # Example usage
40 inv=Inventory()
41 inv.add_item("Pen",10)
42 inv.remove_item("Pen",5)
43 print(inv.get_stock("Pen"))
44
45 # Test is_anagram function
46 print(is_anagram("listen","silent"))
47 print(is_anagram("hello","world"))
48 print(is_anagram("a","a"))
49 print(is_anagram("",""))
50 print(is_anagram("a","b"))
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331 print(is_anagram("listen","silent"))
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333 print(is_anagram("a","a"))
334 print(is_anagram("",""))
335 print
```

(Inventory Class – Apply AI to Simulate Real- World Inventory System)

- Task: Ask AI to generate at least 3 assert-based tests for an Inventory class with stock management.
- Methods:
 - o add_item(name, quantity)
 - o remove_item(name, quantity)

```
o get_stock(name)
```

Example Assert Test Cases:

```
inv = Inventory()
```

```
inv.add_item("Pen", 10)
```

```
assert inv.get_stock("Pen") == 10
```

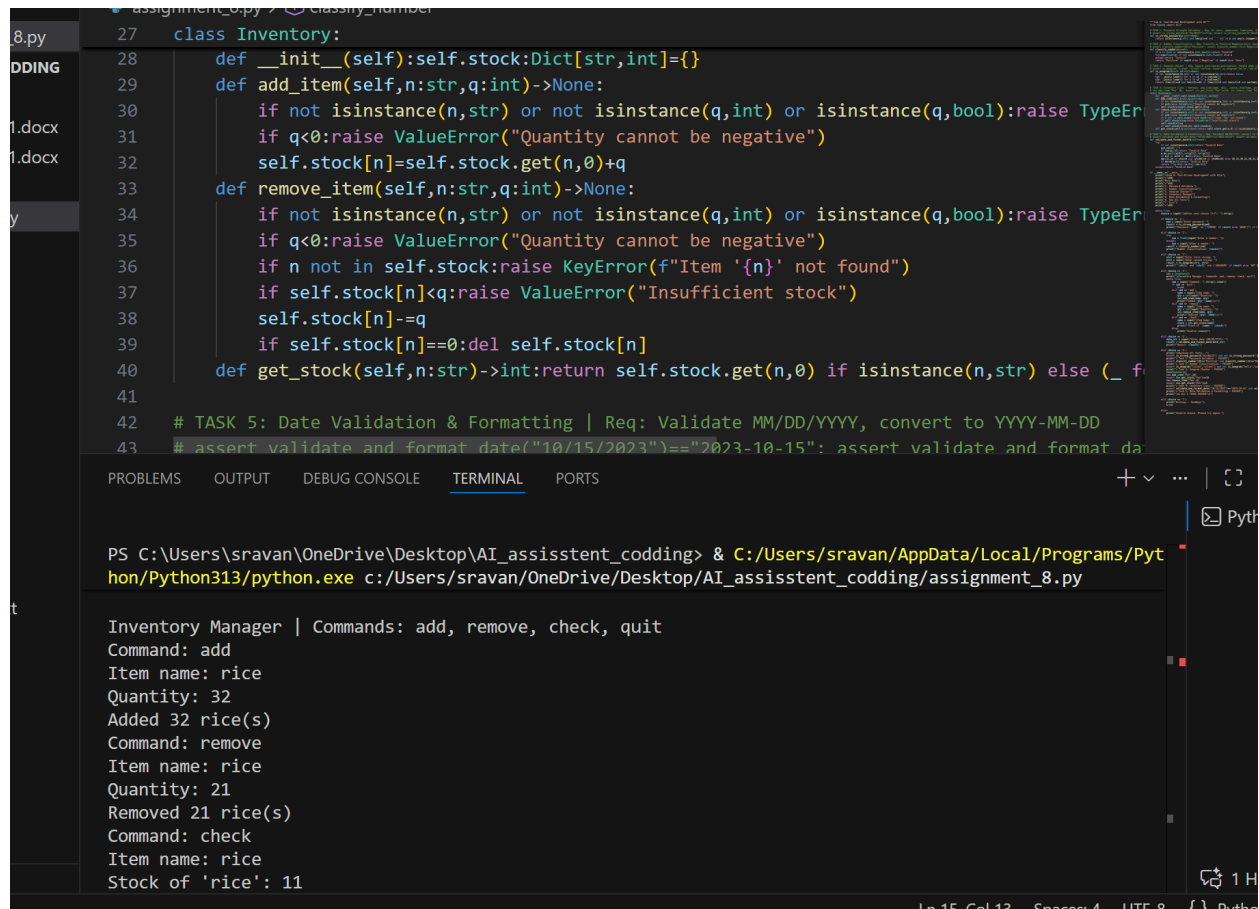
```
inv.remove_item("Pen", 5)
```

```
assert inv.get_stock("Pen") == 5
```

```
inv.add_item("Book", 3)
```

```
assert inv.get_stock("Book") == 3
```

Code:



```
27 class Inventory:
28     def __init__(self):self.stock:Dict[str,int]={}
29     def add_item(self,n:str,q:int)->None:
30         if not isinstance(n,str) or not isinstance(q,int) or isinstance(q,bool):raise TypeError
31         if q<0:raise ValueError("Quantity cannot be negative")
32         self.stock[n]=self.stock.get(n,0)+q
33     def remove_item(self,n:str,q:int)->None:
34         if not isinstance(n,str) or not isinstance(q,int) or isinstance(q,bool):raise TypeError
35         if q<0:raise ValueError("Quantity cannot be negative")
36         if n not in self.stock:raise KeyError(f"Item '{n}' not found")
37         if self.stock[n]<q:raise ValueError("Insufficient stock")
38         self.stock[n]-=q
39         if self.stock[n]==0:del self.stock[n]
40     def get_stock(self,n:str)->int:return self.stock.get(n,0) if isinstance(n,str) else (_ f
41
42 # TASK 5: Date Validation & Formatting | Req: Validate MM/DD/YYYY, convert to YYYY-MM-DD
43 # assert validate and format date("10/15/2023")=="2023-10-15": assert validate and format da

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\sra\OneDrive\Desktop\AI_asisstent_coding> & C:/Users/sra/AppData/Local/Programs/Python/Python313/python.exe c:/Users/sra/OneDrive/Desktop/AI_asisstent_coding/assignment_8.py

Inventory Manager | Commands: add, remove, check, quit
Command: add
Item name: rice
Quantity: 32
Added 32 rice(s)
Command: remove
Item name: rice
Quantity: 21
Removed 21 rice(s)
Command: check
Item name: rice
Stock of 'rice': 11
```

Task Description #5

(Date Validation & Formatting – Apply AI for Data Validation)

- Task: Use AI to generate at least 3 assert test cases for `validate_and_format_date(date_str)` to check and convert dates.

- Requirements:

- o Validate "MM/DD/YYYY" format.

- o Handle invalid dates.

- o Convert valid dates to "YYYY-MM-DD".

Example Assert Test Cases:

```
assert validate_and_format_date("10/15/2023") == "2023-10-15"
```

```
assert validate_and_format_date("02/30/2023") == "Invalid Date"
```

```
assert validate_and_format_date("01/01/2024") == "2024-01-01"
```

Code:

```
assignment_8.py > classify_number
41
42 # TASK 5: Date Validation & Formatting | Req: Validate MM/DD/YYYY, convert to YYYY-MM-DD
43 # assert validate_and_format_date("10/15/2023")== "2023-10-15"; assert validate_and_format_da
44 def validate_and_format_date(d:str)->str:
45     try:
46         if not isinstance(d,str):return "Invalid Date"
47         p=d.split('/')
48         if len(p)!=3:return "Invalid Date"
49         m,da,y=int(p[0]),int(p[1]),int(p[2])
50         if m<1 or m>12 or da<1:return "Invalid Date"
51         dm=[31,29 if y%4==0 and (y%100!=0 or y%400==0) else 28,31,30,31,30,31,31,30,31,30,31
52         if da>dm[m-1]:return "Invalid Date"
53         return f"{y:04d}-{m:02d}-{da:02d}"
54     except:return "Invalid Date"
55
56 if __name__=="__main__":
57     print("\nLab 8: Test-Driven Development with AI\n")
58     print("="*50)
59     print("Main Menu")
60     print("="*50)
61     print("1. Password Validator")

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
+ v ...

Enter your choice (1-7): 5
Enter date (MM/DD/YYYY): 12122004
Result: Invalid Date

Enter your choice (1-7): 5
Enter date (MM/DD/YYYY): 01122026
Result: Invalid Date

Enter your choice (1-7):
```

